Folk-Style Classification (15 different classes) on Mel-Spectrograms using pre-trained models.

1. **My Approach :** Fifteen regional folk styles (e.g. Bauls, Garba, Lavani, Sufi, Bhatiali) are treated as an image classification problem by feeding single-channel spectrogram trains into a pretrained ResNet-18, ResNet-50, Alexnet etc.

2. **Related work :**

Title	Date	Dataset	Model used	Accuracy
An improved ViT model for music genre classification based on mel spectrogram (Wu et al.)	2025	GTZAN (10 genres)	Improved ViT4 (CNN + Transformer hybrid + Enhanced Channel Attention)	86.6%
Music Genre Classification Using Prosodic, Stylistic, Syntactic and Sentiment-Based Features (Kovacs & Baghiu)	2025	14,000 Romanian songs (tabulaturi.ro), 7 merged genres	Logistic Regression	69.94% mean F1 (test set)
uFoC: regional classification of Turkish folk music	2025	TuFoC dataset: 369 recordings 5 Turkish regions	CNN (MobileNetV2, transfer learning)	92.68%
Musical Genre Classification Using Mel-Spectrograms and Mel-Scalograms	2025	GTZAN (10 genres)	MobileNetV2 EfficientNetB0 ResNet50 VGG16	76.7 ± 4.9% 83.5 ± 2.2% 83.3 ± 1.5% 74.3 ± 3.6%

Improved AST paper (To be done.) : PaSST (2022)

3. **Dataset :** Indian Folk Music Dataset (Anupam Biswas sir) contains only .pickle file
   Link : https://zenodo.org/records/6584021

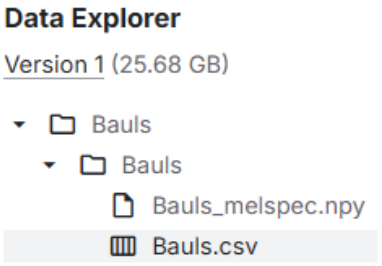
4. **Data-Preprocessing :** Each (.pickle) file consists of two different things - mel-spectrograms (the actual sound) and the meta-data(genre, song, artist etc).

   Every 3-second audio segment is converted into a mel-spectrogram — a time-frequency representation that captures how energy is distributed across perceptually-spaced frequency bands (mel bins) over time. Each spectrogram has a nominal shape of 128 mel-bins by approximately 130 time frames**.**

   For Mel-spectrogram :
   1. Length standardization (fix_length): Forces every spectrogram to have the exact same number of time frames (columns) (128x130).

2.Normalization (standardize): eTakes a spectrogram and rescales its values so they have a mean of 0 and a standard deviation of 1 .
3.SpecAugment-style augmentation (spec_augment): A data augmentation trick — it randomly "blacks out" small chunks of the spectrogram during training, to make the model more robust .

Kaggle link
https://www.kaggle.com/datasets/mishbhaul/pre-processed-indian-folk-music
Each Folk class looks like this :



5. Data Visualization :

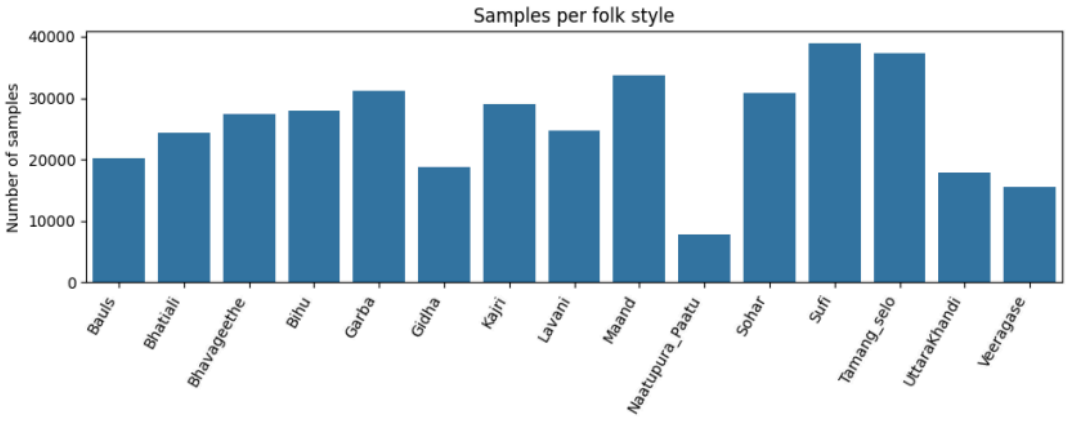


Figure 1. Sample counts per folk style

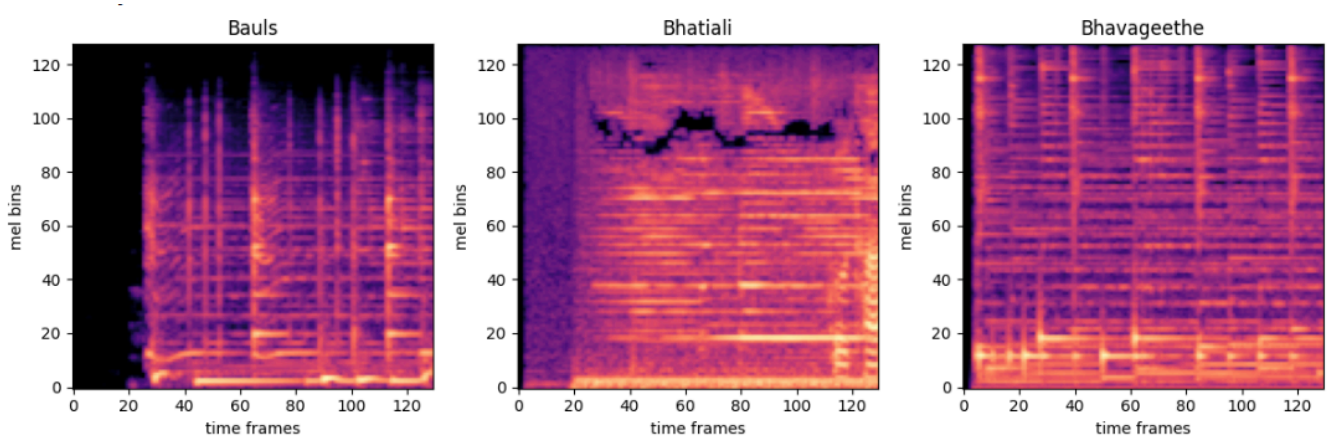


Figure 2. (3 out of 15 classes) mel-spectrograms

5. Challenges Faced :

(a) *Preventing data leakage*: To fix this, splitting is done by song, not by segment — using `GroupShuffleSplit` — so every segment from a given song goes entirely into either training or validation, never both. This guarantees the validation set only contains songs the model has truly never seen before.

(b) *Inefficient raw storage format:*

- Pickle files mixed large spectrogram arrays with small metadata, making them slow and clunky to load
- Fixed by splitting them apart: spectrograms saved as `.npy` files, metadata saved as `.csv` files
- One `.npy` + `.csv` pair per folk style

6. Experimental Setup :

Environment	Kaggle and Google Colab notebooks, single GPU (T4-class)
Data Split	Song-grouped hold-out validation split via `GroupShuffleSplit`, with a fixed random seed for reproducibility across runs.
Batch size	16 (for every models except AST)
Learning Rate	1e-4
Evaluation protocol	`classification_report` (precision, recall, F1 per folk style, macro/weighted averages) + confusion matrix

7. Results & performance Comparison :

All scores are macro avg. of each class.

Model	Val. Accuracy	Precision	Recall	F1-score
AlexNet	0.73	0.70	0.69	0.69
ResNet-34	0.75	0.71	0.70	0.70
ResNet-18	0.762	0.72	0.72	0.71
MobileNet	0.75	0.71	0.70	0.70
EfficientNet-b0 (SOTA)	0.76	0.73	0.72	0.71
AST(SOTA)				

8. Confusion Matrix : (EfficientNet)

